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GR Letter

Paleomagnetic evidence for a stationary Marion hotspot: Additional paleomagnetic data from Madagascar

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Abstract

The island of Madagascar experienced widespread magmatism at ca. 90 Ma due to its interaction with the Marion hotspot. Previous paleomagnetic data from igneous rocks in the southwestern and northwestern regions of the island indicated that the Marion hotspot has remained fixed for the past 90 Ma. We report paleomagnetic data from northeastern Madagascar (d'Analava Complex). Samples were collected from basalts, rhyolites, gabbros and a dolerite dyke. Sixty samples from 5 sites yield a paleomagnetic pole at 66.7°S, 43.5°E (A95=10.7°) and a grand mean pole (GMP) calculated from 10 different studies covering the entire island of Madagascar falls at 68.9°S, 49.0°E (A95=4.4°). This pole translates to a paleolatitude for the Volcan de l'Androy (focal point of the hotspot) at 45.2°+6°/−5°S compared to the current location of the Marion hotspot at ~46°S. Our results confirm, and expand upon, previous studies that argue for the fixity of the Marion hotspot for the past 90 Ma.
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Keywords: Marion hotspot; Paleomagnetism; Madagascar; Cretaceous; Paleolatitude

1. Introduction

The breakup of the eastern Gondwana continent within Pangea commenced in mid-Jurassic times with the initiation of rifting between the combined Madagascar–India blocks at around 165 Ma (see Coffin and Rabinowitz, 1988; Fig. 1a). Torsvik et al. (1998) argued that following the cessation of seafloor spreading between Africa and Madagascar at 130 Ma, widespread magmatism ensued due to the interaction of Madagascar and the Marion hotspot. They further argued that the paleomagnetic data were consistent with the fixity of the Marion hotspot for the last 90 Ma. In a follow-up study, Riisager et al. (2001) collected 144 drill cores from 15 sites along the northwestern part of the island of Madagascar (Fig. 1b) and published a detailed paleomagnetic pole for the Late Cretaceous of Madagascar with coordinates at 67.8°S, 48.5°E). Their study also agreed with the conclusions of Torsvik et al. (1998) regarding the fixity of the Marion hotspot. Since the igneous activity took place during the Cretaceous Normal Superchron (CNS), Riisager et al. (2001) also provided an estimate for the intensity of the paleofield and an analysis of paleosecular variation. The conclusion was that paleosecular variation recorded in these eruptive products was somewhat lower than expected, but not statistically significant.

In an effort to complete the island-wide sampling of these Late Cretaceous eruptive products, we collected 5 additional paleomagnetic sites from northeastern Madagascar where the d'Analava gabbroic Complex (91.6±3 Ma; Torsvik et al., 1998) is thought to represent one of the feeder magma chambers to the Late Cretaceous volcanic rocks. We report the results in this paper and derive a grand mean pole (GMP) for the island of Madagascar.

2. Sampling procedure

Fig. 1b shows the locations of paleomagnetic sampling sites in the Late Cretaceous igneous rocks and Table 1 gives the GPS coordinates of the sampling locations. We collected 70 samples from 5 sites. One of our sites included a doleritic dike that intruded country rocks thought to have undergone metamorphism during the Late Neoproterozoic (Meert et al., in

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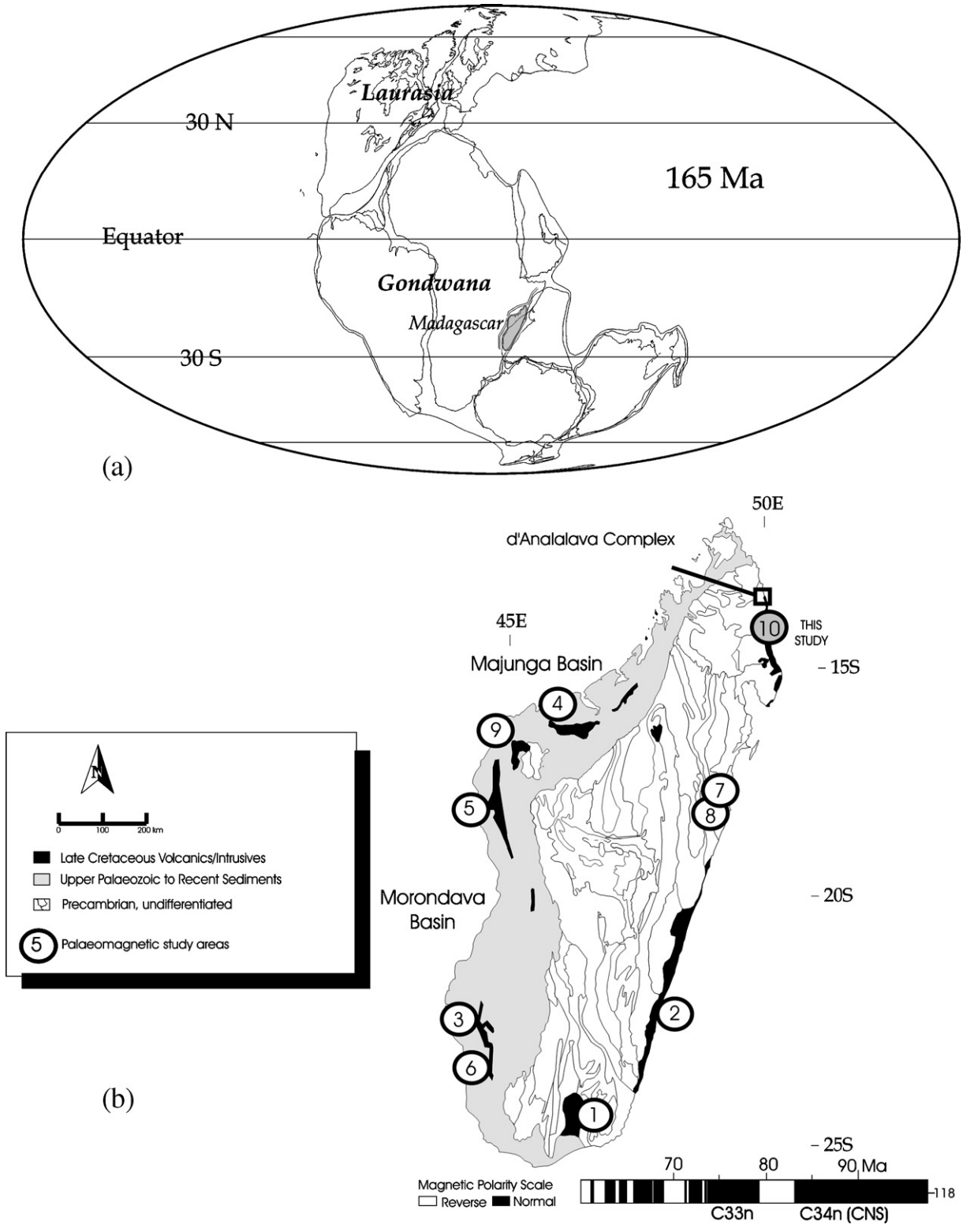


Fig. 1. (a) The supercontinent Pangea during the Middle Jurassic 175 Ma just prior to breakup. (b) The island of Madagascar showing locations of previous paleomagnetic studies (1–9) and this study (10) along with a generalized geological map (after Torsvik et al., 1998).

preparation). Two sites were from basalts, one site was collected from rhyolites and samples from the last site were taken from the d'Analava gabbroic complex.

All samples were drilled in the field using a gasoline-powered drill and oriented using both magnetic and sun compasses. The samples were then cut into specimens at the University of Florida. Individual samples were then stepwise treated using thermal (60 min at each temperature level) or alternating field demagnetization and a treatment sequence was selected that resulted in the clearest separation of

Table 1
Paleomagnetic results

Site	Rock Type	Dec	Inc	*k*	α95	*N*	Plat (°S)	Plong (°E)	dp	dm
1. CA, 14.27°S, 50.16°E	Basalt	7.2	−50	157	2.5	21	75.5	26.1	2.1	3.3
2. KG, 13.57°S, 49.98°E	Gabbro	7.8	−49	198	3.9	8	72.5	26.8	3.4	5.1
3. DI, 13.64°S, 49.76°E	Dolerite+baked host rock	2.1	−45.4	77	5.5	10	76.6	41.6	4.4	7.0
	Unbaked	273.6	+68.8	81	10.2	4	8.6	11.5	14.7	17.3
4. KR, 14.20°S 50.05°E	Rhyolite	1.5	−67.1	81	5.1	11	54.4	48.4	7	8.5
5. KB, 14.45°S, 50.17°E	Basalt	355.6	−68.8	135	4.2	10	52.1	54.6	6	7.1
Overall		3.6	−56.1	52	10.7	60	66.7	43.5		

Dec=declination, Inc=inclination, *k*=kappa precision parameter (Fisher, 1953), α95=cone of 95% confidence (Fisher, 1953), *N*=number of samples used, Plat=latitude of the paleomagnetic pole, Plong=longitude of the paleomagnetic pole, dp, dm=cone of 95% confidence about the paleomagnetic pole in the co-latitude direction (dp) and at a right angle to the co-latitude direction (dm).

individual vector components. All samples were stored in a μ-metal shield for 1 month prior to measurement. Alternating field demagnetization was conducted using a home-built AF-demagnetizer able to reach peak fields of 140 mT and thermal demagnetization was conducted using an ASC-Scientific thermal demagnetizer. Samples were measured in a Molspin® spinner magnetometer.

Curie temperature runs were conducted on powdered samples using the KLY-3S susceptibility bridge adapted with a CS-3 heating unit (AGICO). Susceptibility measurements are made incrementally during the heating and cooling of the samples. In an effort to minimize chemical alteration of the samples during heating, a constant flow of high purity argon gas was pumped into the system at a constant flow rate (100 ml min^{-1}).

3. Results

3.1. d'Analava gabbro

A total of 8 samples were collected from the gabbroic body at d'Analava (13.57°S; 49.98°E). Fig. 2(a–f) shows typical demagnetization behavior of the gabbro samples using both alternating field (Fig. 2a–c) and thermal demagnetization (Fig. 2d–f). Alternating field demagnetization up to peak fields of 140 mT was only able to remove just over 50% of the natural remanent magnetization (NRM) in most samples (Fig. 2a–c). Thermal demagnetization (Fig. 2d–f) resulted in a much clearer demagnetization trajectory with a discrete unblocking temperature range between 500 and 575 °C and the remaining samples were thermally treated. The overall mean direction (after cleaning) was declination (D=7.8°) and inclination (I=−49°, *k*=198 and α95=3.9°, Fig. 2g) with a resultant virtual geomagnetic pole (VGP) at 72.5°S, 26.8°E (dp=3.4°; dm=5.1°). Given that Madagascar was in the southern hemisphere during the emplacement of the gabbro, the polarity of the samples is all normal. The age of the d'Analava complex was reported by Torsvik et al. (1998) at 91.6±3 Ma. Due to the discrete unblocking temperature spectra and the relatively high coercivity of the samples, we conclude that low-Ti magnetite is the main carrier. Curie temperature runs (Fig. 2h) also show a very sharp drop in susceptibility at 578 °C during the heating phase and some slight alteration of the magnetite grains upon cooling. These are both consistent with relatively low-Ti magnetite as the dominant carrier in the samples.

3.2. Cretaceous basalts

A total of 31 samples were collected from the basaltic flows at 2 different sites (14.27°S; 50.16°E and 14.45°S, 50.17°E). Fig. 3(a–f) shows typical demagnetization behavior of the basaltic samples using both alternating field (Fig. 3a–c) and thermal demagnetization (Fig. 3d–f). Alternating field demagnetization up to peak fields of 100 mT was usually able to remove more than 80% of the natural remanent magnetization (NRM) in most samples (Fig. 3a–c). Thermal demagnetization (Fig. 3d–f) resulted in a clean demagnetization trajectory with a distributed unblocking range from ~ 200 to 525 °C. Half of the remaining samples were thermally demagnetized and the other half was demagnetized using alternating field methods. The overall mean direction (after cleaning) for the first basalt site (CA-samples) was declination (D=7.2°) and inclination (I=−50°) with *k*=157 and α95=2.5° with a resultant virtual geomagnetic pole (VGP) at 75.5°S, 26.1°E (dp=2.1°; dm=3.3°). These basalts are thought to be genetically related to the 91.6 Ma d'Analava gabbro complex. Due to the relatively low and distributed unblocking temperature spectra and the relatively low coercivity of the samples, we conclude that the carrier of the magnetization in the basalts is titanomagnetite. Curie temperature runs (Fig. 3h) also show a very sharp drop in susceptibility at 578 °C during the heating phase and some slight alteration of the magnetite grains upon cooling.

3.3. Cretaceous rhyolite

A total of 11 samples were collected from a small outcrop of rhyolite outside of the town of Sambava, Madagascar (14.20°S; 50.05°E). Fig. 4(a–f) shows typical demagnetization behavior of the rhyolite samples using both alternating field (Fig. 4a–c) and thermal demagnetization (Fig. 4d–f). Alternating field demagnetization up to peak fields of 140 mT was usually only able to remove just over 50% of the natural remanent magnetization (NRM) in most samples (Fig. 4a–c). Thermal demagnetization (Fig. 4d–f) resulted in a slightly better demagnetization trajectory with distributed unblocking temperatures range between 300 and 590 °C, but

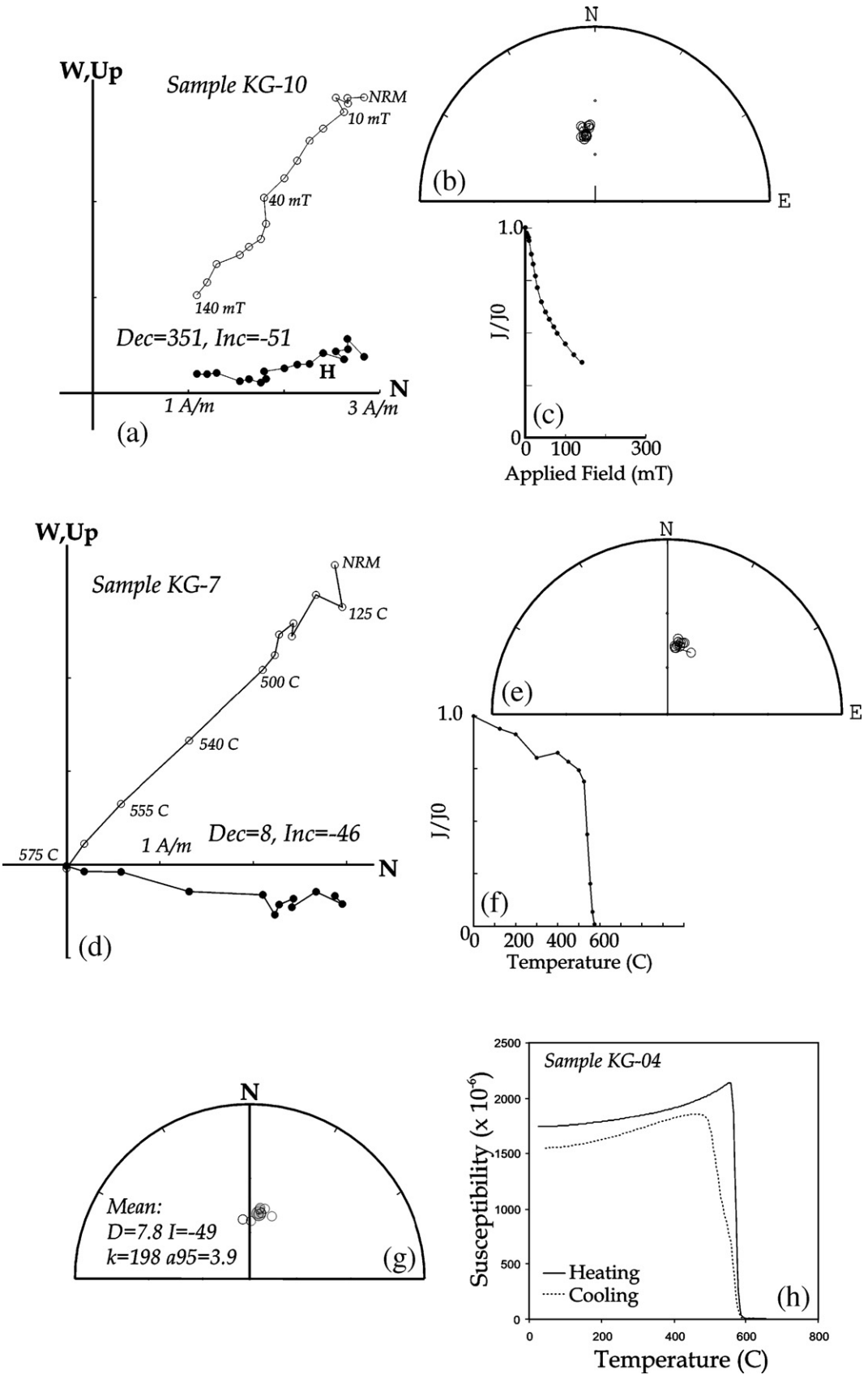


Fig. 2. Paleomagnetic results from the 90 Ma d'Analava gabbro; (a) alternating field demagnetization vector diagram for sample KG-10; (b) stereoplot of the demagnetization vectors for sample KG-10; (c) intensity decay plot for KG-10; (d) thermal demagnetization vector diagram for sample KG-7; (e) stereoplot of the demagnetization vectors for sample KG-7; (f) intensity decay plot of sample KG-7; (g) stereoplot of the mean direction obtained from the d'Analava gabbro samples and (h) curie temperature run of sample KG-4 indicating magnetite as the main magnetic mineral in the sample.

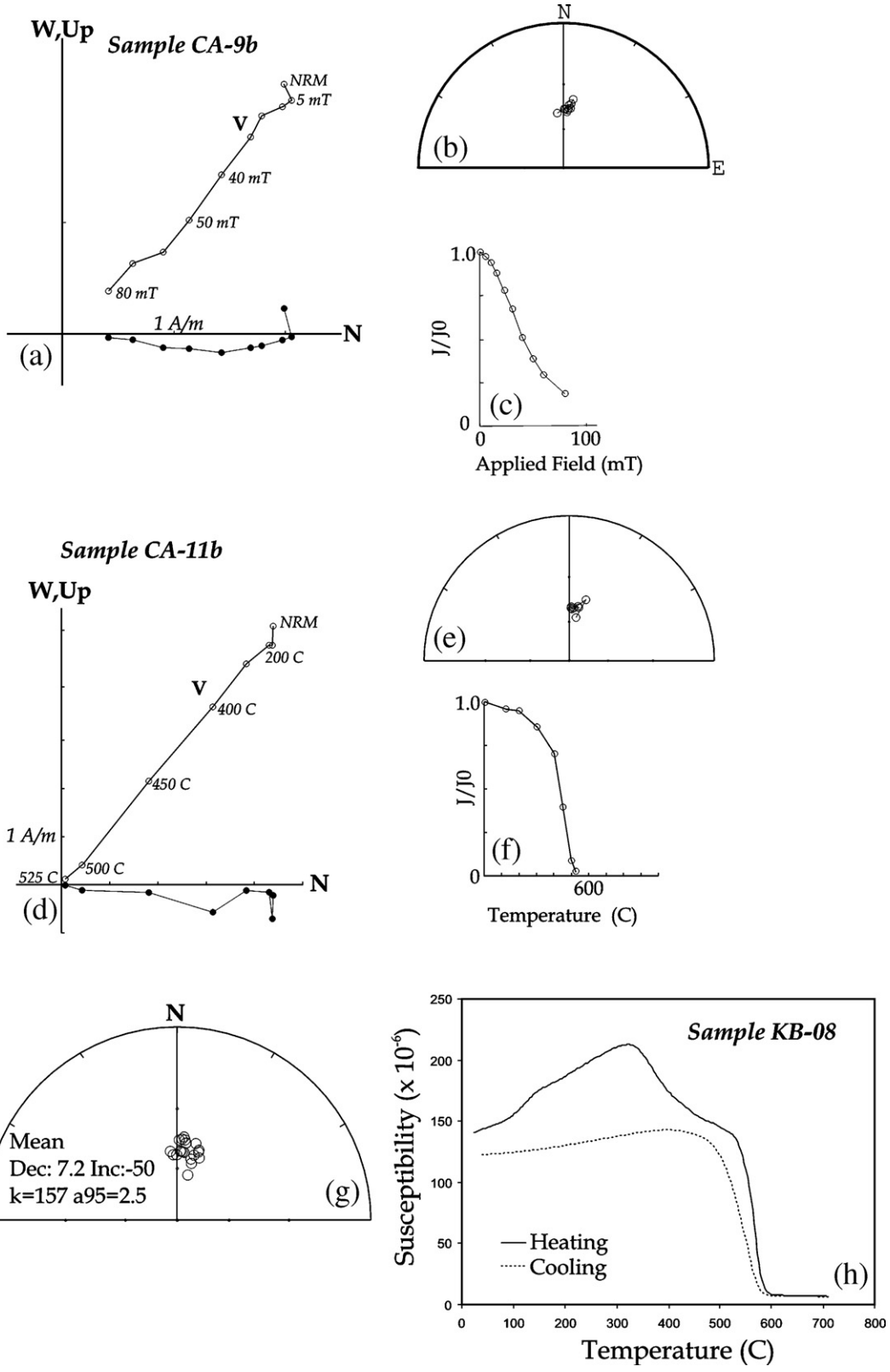


Fig. 3. Paleomagnetic results from the 90 Ma d'Analava basalts; (a) alternating field demagnetization vector diagram for sample Ca-9b; (b) stereoplot of the demagnetization vectors for sample CA-9b; (c) intensity decay plot for CA-9b; (d) thermal demagnetization vector diagram for sample CA-11b; (e) stereoplot of the demagnetization vectors for sample CA-11b; (f) intensity decay plot of sample CA-11b; (g) stereoplot of the mean direction obtained from the d'Analava basalt samples and (h) curie temperature run of sample KB-8 indicating magnetite and a secondary magnetic phase (pyrrhotite) as the main magnetic minerals in the sample.

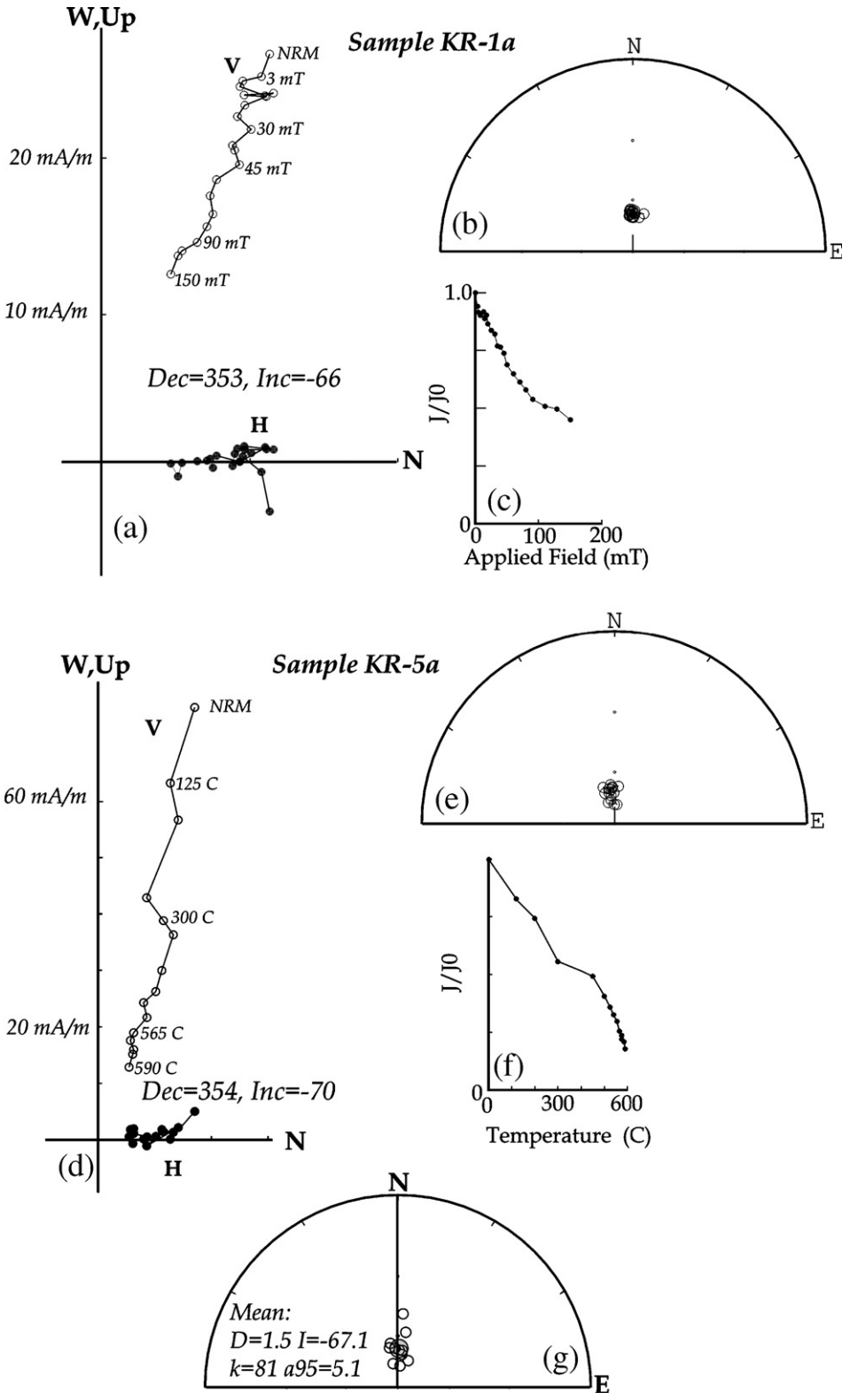


Fig. 4. Paleomagnetic results from the d'Analava rhyolites; (a) alternating field demagnetization vector diagram for sample KR-1a; (b) stereoplot of the demagnetization vectors for sample KR-1a; (c) intensity decay plot for Kr-1a; (d) thermal demagnetization vector diagram for sample KR-5a; (e) stereoplot of the demagnetization vectors for sample KR-5a; (f) intensity decay plot of sample KR-5a; (g) stereoplot of the mean direction obtained from the d'Analava rhyolite samples.

heating to 590 °C did not completely remove the magnetization. Several samples were heated to 690 °C but their behavior became erratic at temperatures above 600°. Half of the remaining samples were treated thermally and the other half using alternating field. The overall mean direction (after cleaning) was declination (D=1.5°) and inclination (I=

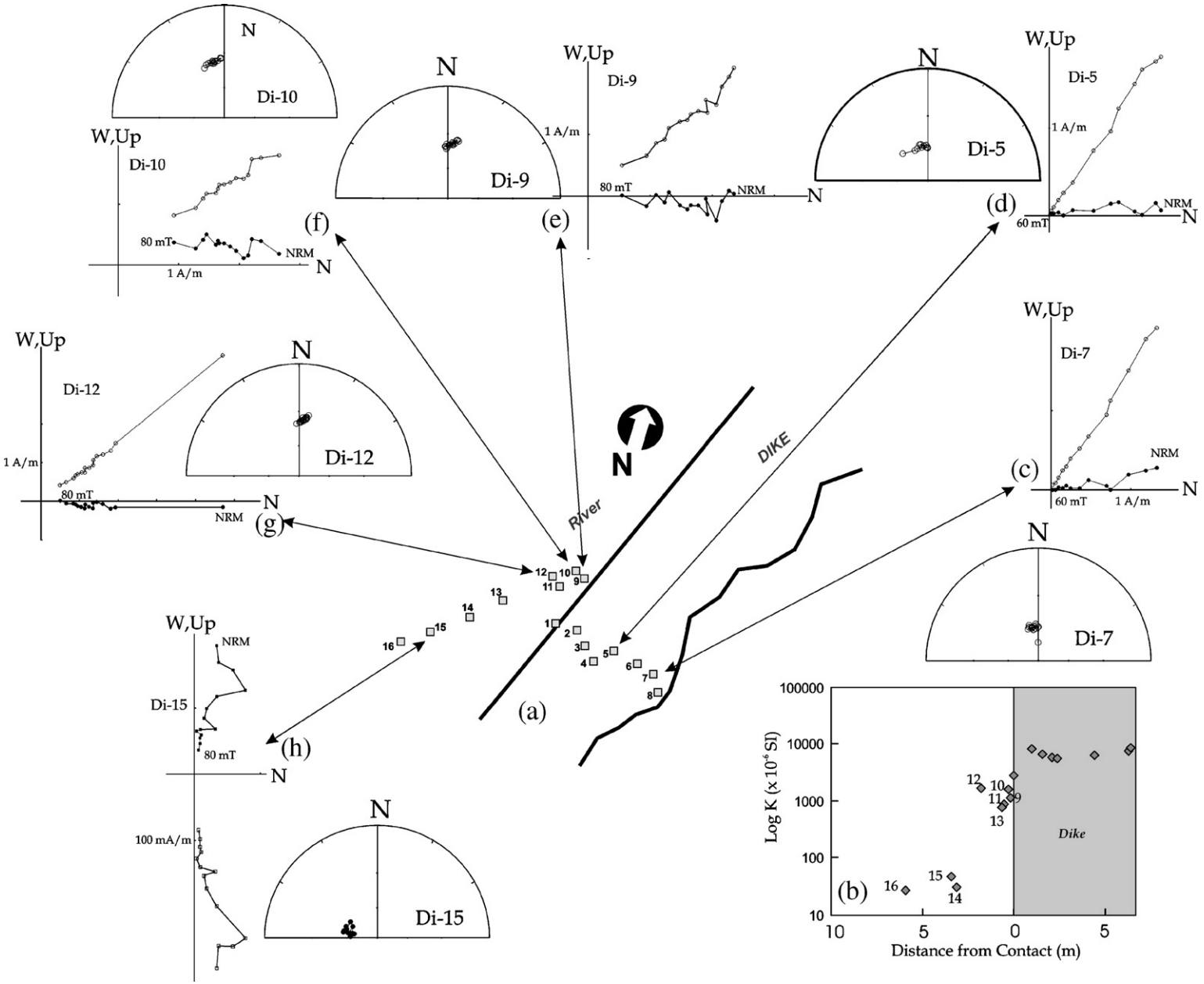


Fig. 5. Paleomagnetic results from the Milanoa dike and country rock; (a) conceptual map showing the sampling locations of dike and country rock; (b) plot of log susceptibility versus distance from the dike contact; (c) alternating field demagnetization diagram for dike sample DI-7 and a stereoplot of the vectors; (d) alternating field demagnetization diagram for dike sample DI-5 and a stereoplot of the vectors; (e) alternating field demagnetization diagram for country rock sample DI-9 and a stereoplot of the vectors; (f) alternating field demagnetization diagram for country rock sample DI-10 and a stereoplot of the vectors; (g) alternating field demagnetization diagram for country rock sample DI-12 and a stereoplot of the vectors and (h) alternating field demagnetization diagram for country rock sample DI-15 and a stereoplot of the vectors.

Table 2
Summary of paleomagnetic studies

Location	Pole lat (°S)	Pole long (°E)	A95	Reference
1. Massif d'Androy	64	63	7.7	Andriamirado, 1971
2. SE Coast	65.8	35.6	4.4	Andriamirado, 1971
3. Mangoky-Anilahy	73.7	73.1	10	Andriamirado, 1971
4. Antanimena	66.1	79.7	4.9	Andriamirado, 1971
5. Mailaka	70.3	63.1	6.9	Andriamirado, 1971
6. SW Madagascar	76.8	68.2	2.4	Torsvik et al., 1998
7. Tamatave-1	60.2	32.1	2.8	Andriamirado, 1971
8. Tamtave-2	65.5	38	6.0	Storetvedt et al., 1992
9. Antanimena-Malaika	74	43.7	5.3	Riisager et al., 2001
10. d'Analava Complex	66.7	43.5	12	This study
Mean 1–10	68.9	49.0	4.4	This compilation

−67.1°, k=81 and α95=5.1°, Fig. 4g) with a resultant virtual geomagnetic pole (VGP) at 54.4°S, 48.4°E (dp=7.0°; dm=8.5°). The age of the rhyolites is also thought to be related to magmatic activity associated with the 91.6 Ma d'Analava complex (see Torsvik et al., 1998). Due to the discrete unblocking temperature spectra and the relatively high coercivity of the samples, we conclude that low-Ti magnetite is the main carrier.

3.4. Cretaceous dike

A 6.6 m wide dike located along the Milanoa River (13.64°S; 49.76°E) intruded high grade metamorphic rocks of unknown age. The dike shows very sharp chilled margins consistent with emplacement into relatively cool country rock. A $^{40}Ar/^{39}Ar$ geochronologic study on the same metamorphic rocks (4 km south; Meert et al., in preparation) yielded plateau ages of 562 ±5 and 556±1 Ma. Other metamorphic rocks in the vicinity of d'Analava yield $^{40}Ar/^{39}Ar$ biotite ages ranging from 478 Ma to 555 Ma (Meert et al., in preparation) A total of 16 samples were taken from the dike and host rocks. Susceptibility measurements were made on each core using a Sapphire Instruments (SI-2B) bridge and plotted as a function of distance from the contact (Fig. 5a). Susceptibility values are orders of magnitude higher in the dike and the country rock immediately adjacent to the dike than in the unbaked country rocks. Samples were treated using alternating field demagnetization and the results are shown in Fig. 5b–g. Samples from the dike and country rock within 1/2 dike width of the margin are all of normal polarity although samples from the dike tend to be somewhat steeper in inclination than the baked host (see Table 2). Samples from the unbaked host show consistent westerly-steeply down directions that are quite distinct from the dike. This provides strong evidence for a positive baked contact test. The resultant VGP

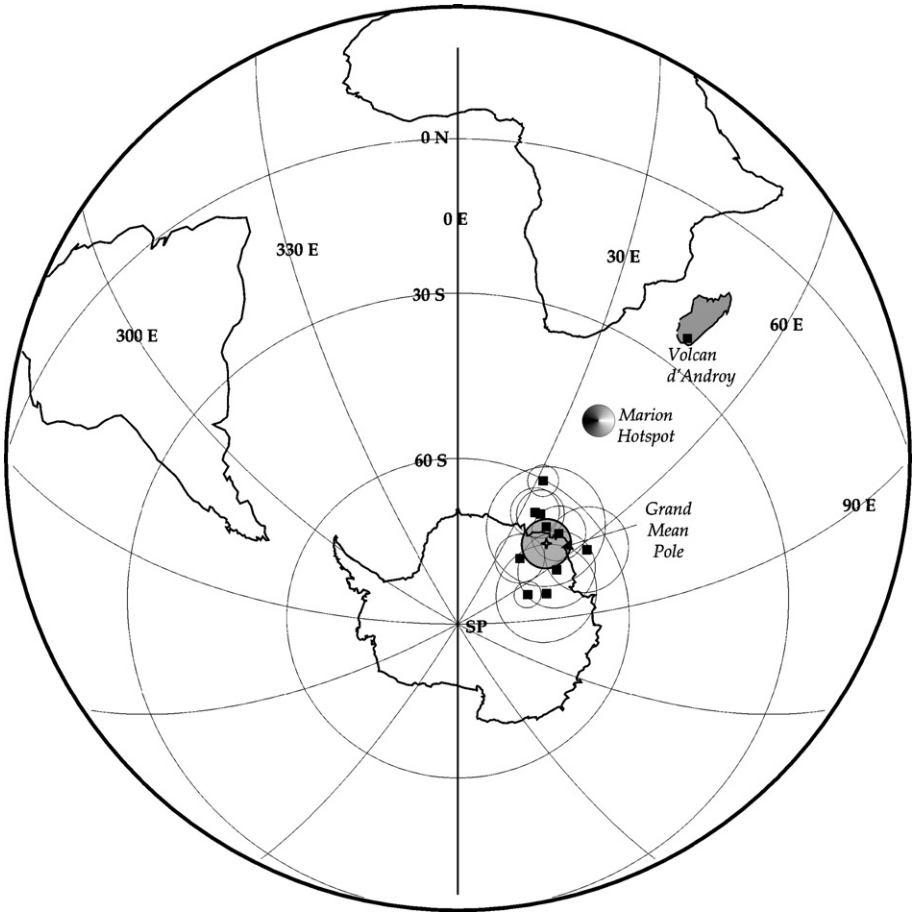


Fig. 6. Compilation of previous paleomagnetic studies and location of the grand mean pole at 68.9°S, 49°E (A95=4.4°, Q=7). Also shown are the locations of the Marion hotspot and the Volcan d'Androy that is thought to be the focal point for the Marion hotspot volcanic activity at ca. 90 Ma.

from the dike and the baked contact rocks falls at 76.6°S, 41.6°E (dp=4.4°, dm=7.0°). The VGP from the unbaked host rocks falls at 8.6°S, 11.5°E. This VGP falls near other Cambrian poles from Madagascar (Meert et al., 2001, 2003).

4. Discussion and conclusions

A paleomagnetic pole calculated from these 5 sites falls at 66.7°S, 43.5°E (A95=10.7°). If we average our pole with the summary of paleomagnetic poles compiled by Riisager et al. (2001), we arrive at a grand mean pole (Table 2; Fig. 6) at 68.9°S, 49°E (A95=4.4°, Q=7). The focal point of the Marion plume is hypothesized to be at 24.2°S, 46°E (Volcan de l'Androy). The paleolatitude calculated for this location using the grand mean pole is $45.2°^{+6°}_{-5°}$. The present-day latitude of the Marion hotspot is ~46°S. Thus, our study completes the examination of Cretaceous volcanism from around the island with a sampling of the d'Analava complex in northeastern Madagascar and confirms earlier results arguing for the fixity of the Marion hotspot for the last 90 Ma (Fig. 6; Torsvik et al., 1998; Riisager et al., 2001).

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